Graphical Abstract

Scatter-Regularized Cross-Multi-Kernel Contrastive Learning for One-Class Anomaly Detection

Highlights

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- A cross-multi-kernel contrastive encoder reconciles multi-scale kernel views by treating the kernel index as the contrastive view.
- A multi-kernel scatter regularizer is proved equal to the average within-class scatter and to one-class centroid multiple-kernel k-means.
- A dual-signal detector fuses an angular (linear OC-SVM) and a radial (RBF OC-SVM) score, staying robust across datasets where either single signal collapses.

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Abstract

One-class anomaly detection has attracted extensive research due to its broad application in identifying unforeseeable deviations. The idea of contrastive learning is one of the widely used methods in modern outlier analysis. Nevertheless, these methods mainly use cross-view consistency to model normal data, so they cannot effectively guarantee the absolute intra-class compactness crucial for identifying outliers. Moreover, they typically map data into a single, fixed geometric space and cannot effectively capture complex similarity structures across multi-scale data. In this work, we propose a Scatter-regularized Cross-Multi-Kernel (SCMK) contrastive framework for one-class anomaly detection. First, a bank of multi-scale Gaussian kernels is constructed to inject multi-resolution geometric priors. Second, a cross-kernel InfoNCE objective is defined to treat the kernel index as a distinct view and align the representations. Then, a novel multi-kernel scatter regularizer is introduced to explicitly compact the normal class within every kernel-induced space. Based on this idea, two complementary signals–an angular detector (linear OC-SVM) and a radial detector (RBF OC-SVM)–are fused via a per-sample maximum to quantify the anomaly scores. Finally, experimental comparisons with state-of-the-art methods on 20 benchmark datasets show that the proposed method has a significant improvement and applies robustly to diverse dataset topologies.

Keywords: anomaly detection, multiple kernel learning, contrastive learning, one-class classification, representation learning

1. Introduction

Anomaly detection—the task of identifying observations that deviate markedly from the prevailing pattern of the data—underpins a broad range of applications, including financial fraud detection, industrial fault diagnosis, network-intrusion monitoring, and medical screening [13, 14]. The defining difficulty of the problem is statistical: anomalies are rare, heterogeneous, and largely unforeseeable, so the abnormal class can be neither sampled exhaustively nor characterised in advance. This rules out a conventional supervised formulation and motivates the *one-class* (semi-supervised) paradigm, in which a model is trained on normal data alone and flags as anomalous any test instance that departs from the learned notion of normality [1, 2].

Classical one-class detectors realise this idea in a *single, fixed* similarity space. The one-class support vector machine (OC-SVM) [1] and support vector data description (SVDD) [2] separate the normal data from the origin, or enclose them in a minimal sphere, within one reproducing-kernel Hilbert space; density- and distance-based methods such as LOF [3] and isolation forests [4] likewise commit to one geometric prior. Their accuracy therefore hinges critically on the choice of kernel and bandwidth, a choice that does not transfer well across datasets and degrades in high-dimensional, heterogeneous feature spaces. Deep one-class models alleviate the fixed-geometry issue by *learning* a representation: autoencoder-based detectors [7] score samples by reconstruction error, while Deep SVDD [8] maps normal data into a compact hypersphere. Yet reconstruction objectives are not inherently discriminative, hypersphere objectives are prone to representational collapse, and—crucially—each of these methods still summarises similarity through a *single* learned geometry.

Multiple kernel learning (MKL) offers a principled remedy to the single-geometry limitation by combining a bank of kernels that encode complementary notions of similarity at different scales [15, 16]. Fusing such views improves robustness and removes the burden of selecting one "correct" kernel, and the idea extends naturally to clustering through multiple-kernel k-means and its centroid variants [19, 20]. However, classical MKL and MKKM are *shallow*: they operate directly on the raw input features and merely learn a set of scalar kernel weights, without inducing a representation that a downstream detector can exploit. The expressive power of modern representation learning is thus left untapped.

Contrastive representation learning provides exactly such expressive power. By pulling together different "views" of the same instance and pushing apart views of different instances, objectives such as InfoNCE [21] and SimCLR [22] learn embeddings that transfer remarkably well, and they have recently been adapted to anomaly detection [11, 12]. This suggests a natural marriage with MKL: treat the *kernel index itself as a contrastive view*, so that the embeddings produced by different kernels for the same sample are aligned, while embeddings of different samples remain separable. Such a cross-kernel contrastive encoder reconciles multi-scale similarity within a learned representation. Nevertheless, a subtle but consequential gap remains. The InfoNCE objective enforces only *cross-view consistency* (alignment of a sample's kernel views) and *instance discriminability* (uniformity over the sphere); it does not make the normal class *absolutely compact*. Indeed, the loss is invariant to configurations that scatter normal embeddings widely across the representation space as long as each sample's views agree. For one-class detection, however, the compactness of the normal region is precisely the property that a downstream detector relies upon. We identify this mismatch—*consistency is not compactness*—as the central obstacle to coupling contrastive learning with one-class detection.

We resolve it with *scatter-regularized cross-multi-kernel* contrastive learning (SCMK). On top of a cross-kernel contrastive encoder built over a multi-scale Gaussian kernel bank, we introduce a *multi-kernel scatter regularizer* that explicitly compacts the normal class within every kernel-induced embedding space. The regularizer is derived from the one-class specialisation of centroid multiple-kernel *k*-means, and we prove that it equals —up to an additive constant—the average within-class scatter of the unit-norm embeddings, thereby giving the heuristic a firm theoretical footing. Combined with the uniformity pressure of the contrastive term, the scatter penalty tightens the normal class without inducing collapse, yielding a representation on which a simple and efficient *linear* one-class SVM suffices to score anomalies.

The main contributions of this work are summarised as follows.

(1) We propose a cross-multi-kernel contrastive framework that treats the kernel index as the contrastive view and aligns the embeddings of a multi-scale kernel bank whose bandwidths are auto-calibrated by a median-distance heuristic on the normal data.

(2) We introduce a multi-kernel scatter regularizer and *prove* that it is equivalent to the average within-class scatter and to the one-class instance of centroid multiple-kernel *k*-means, bridging multiple-kernel clustering theory and contrastive anomaly detection.

(3) We provide a collapse-avoidance analysis through the alignment–uniformity lens, showing how the contrastive and scatter terms balance, and we exploit the resulting representation with a lightweight dual-signal detector that fuses an angular (linear OC-SVM) and a radial (RBF OC-SVM) score, recovering the discriminative information that ℓ_2 normalization alone discards.

(4) Extensive experiments on twenty benchmark datasets against seven recent detectors demonstrate that the proposed method attains the highest mean and most stable detection accuracy among competitive baselines, and ablation studies isolate the contribution of each component.

The remainder of this paper is organized as follows. Section 2 reviews related work on one-class anomaly detection, multiple kernel learning and contrastive representation learning. Section 3 presents the proposed SCMK method. Section 4 reports the experimental comparison, statistical tests, runtime, parameter sensitivity and ablation studies. Section 5 concludes the paper and outlines future work.

2. Related work

This section organises the literature around the three ingredients that SCMK combines. We first review one-class anomaly detectors and their geometric assumptions, then discuss multiple-kernel learning as a way to encode several similarity scales, and finally examine contrastive representation learning before clarifying the specific gap addressed by our method.

2.1. One-class and deep anomaly detection

Unsupervised anomaly detection has been studied for decades under a variety of paradigms. Traditional approaches can be broadly grouped into statistical, distance-based, density-based and clustering-based methods. Statistical methods fit a model to the bulk of the data and flag observations that deviate significantly from the assumed distribution [5]. Distance-based methods treat points that lie far from most others in the feature space as anomalous [6], whereas density-based methods—of which the Local Outlier Factor (LOF) [3] is the canonical example—compare the local density of a point with that of its neighbours to expose local outliers. Clustering- and partition-based methods instead group the data and regard samples that fall far from any cluster, or that are easily isolated as in the isolation forest [4], as anomalies. Although effective on low-dimensional data, these methods rest on explicit assumptions about the geometry of the normal class—a roughly uniform distribution for distance-based methods, a well-calibrated neighbourhood for density-based methods, or a fixed similarity measure—and consequently degrade on high-dimensional, heterogeneous or categorical data whose structure violates those assumptions.

A particularly influential family formalises detection as *one-class* classification: only normal data are available, and the task is to characterise the compact region they occupy so that deviations can be flagged at test time. The two canonical kernel-based formulations are the One-Class Support Vector Machine (OC-SVM) [1], which separates the normal data from the origin by a maximum-margin hyperplane in a reproducing-kernel Hilbert space, and Support Vector Data Description (SVDD) [2], which encloses the normal data within a minimum-volume hypersphere; the two objectives coincide for translation-invariant kernels. These shallow one-class detectors are effective but remain sensitive to the choice of kernel and similarity measure and scale poorly with dimension, which motivates coupling the one-class objective with a learned representation.

Deep anomaly detection learns the representation jointly with the detector. Autoencoder and robust-autoencoder approaches [7] flag samples with large reconstruction error, and Deep SVDD [8] learns a network that maps normal data into a minimum-volume hypersphere, at the risk of trivial collapse. A complementary line is self-supervised: geometric-transformation prediction [9] and its generalisation to non-image data, GOAD [10], detect anomalies through the predictability of applied transformations.

2.2. Multiple kernel learning

Multiple kernel learning combines several base kernels into a single, more expressive kernel, typically by learning a convex combination of weights. Foundational formulations cast weight learning as semidefinite or gradient-based optimisation [16, 17], and localised variants allow the combination to vary across the input space [18]; see [15] for a survey. In the unsupervised setting, multiple-kernel k-means and its centroid (CMKKM) and matrix-regularised variants [19, 20] jointly optimise cluster assignments and kernel weights, with diversity-promoting regularisers discouraging redundant kernels. These techniques are, however, shallow—they act on raw features and output only kernel weights. Our work retains the multiple-kernel principle but lifts it into a learned embedding space, and it repurposes the one-class CMKKM objective as a differentiable scatter regularizer for representation learning rather than as a clustering criterion.

2.3. Contrastive representation learning

Contrastive learning has become a dominant paradigm for self-supervised representation learning. InfoNCE [21] and SimCLR [22] maximise agreement between augmented views of an instance against a set of negatives, MoCo [23] maintains a momentum queue of negatives, and BYOL [24] and SimSiam [25] dispense with explicit negatives altogether. Wang and Isola [26] characterise the resulting geometry through two competing properties, alignment of positive pairs and uniformity of the embedding distribution on the hypersphere, an analysis we invoke to explain why our scatter regularizer does not collapse the representation. Whereas these methods generate views by stochastic data augmentation, we obtain views from kernel diversity, and we observe that the consistency enforced by contrastive learning does not by itself yield the intra-class compactness required for one-class detection—a gap that our scatter regularizer is designed to close.

2.4. Positioning of this work

SCMK sits at the intersection of the three threads above. From anomaly detection it inherits the one-class formulation and an OC-SVM head; from multiple kernel learning it inherits a multi-scale kernel bank and the centroid-MKKM compactness principle; and from contrastive learning it inherits a representation objective, which we adapt by treating the kernel index as the view. To our knowledge, the combination of kernel-as-view contrastive learning with a scatter regularizer that provably realises one-class CMKKM has not been explored, and it is this combination that delivers a compact, kernel-fused representation amenable to efficient linear one-class detection.

3. The proposed SCMK method

This section develops SCMK from the modelling assumptions to the deployable detector. After formalising the one-class setting, we construct the multi-scale kernel bank, use it to define cross-kernel contrastive embeddings, add the scatter regularizer that enforces compactness, and finish with the dual-signal OC-SVM scoring rule and computational summary.

3.1. Problem formulation and notation

Let $\mathcal{X} = \{\boldsymbol{x}_i\}_{i=1}^{N}$ with $\boldsymbol{x}_i \in \mathbb{R}^D$ denote the data after preprocessing,[1] and let $y_i \in \{0, 1\}$ be the latent label, where $y_i = 0$ marks a normal instance and $y_i = 1$ an anomaly. Only the normal subset $\mathcal{X}_\mathrm{n} = \{\boldsymbol{x}_i : y_i = 0\}$, of size $N_\mathrm{n} = |\mathcal{X}_\mathrm{n}|$, is available at training time; anomalies are neither observed nor simulated. The goal is to learn a scoring function $a : \mathbb{R}^D \to \mathbb{R}$ such that $a(\boldsymbol{x})$ is large for anomalies and small for normal points.

Our framework, termed *scatter-regularized cross-multi-kernel* contrastive learning (SCMK), couples a multiple-kernel contrastive encoder with a one-class detection head. It comprises four components, summarised in Algorithm 1: (i) a multi-scale kernel bank that injects geometric priors at several resolutions (Section 3.2); (ii)

[1] Nominal attributes are one-hot encoded and numerical attributes are min–max scaled to $[0, 1]$, so that all feature dimensions share a comparable range.

a cross-kernel contrastive objective that aligns the views produced by different kernels (Section 3.3); (iii) a multi-kernel scatter regularizer that explicitly compacts the normal class in every kernel-induced embedding (Section 3.4); and (iv) a dual-signal one-class detector that fuses an angular and a radial OC-SVM score to flag anomalies (Section 3.6).

3.2. Multi-scale kernel bank

To endow the encoder with multi-resolution sensitivity, we instantiate a bank of K Gaussian kernels whose bandwidths are anchored to the intrinsic scale of the normal data. Let

$$\text{med} = \text{median}\{ \left\| \boldsymbol{x}_i - \boldsymbol{x}_j \right\|_2 : \boldsymbol{x}_i, \boldsymbol{x}_j \in \mathcal{X}_{\text{n}},\ i \neq j\} \tag{1}$$

be the median pairwise Euclidean distance estimated *from normal samples only*, which prevents contamination of the scale statistic by outliers. The k-th kernel is

$$\kappa_k(\boldsymbol{a}, \boldsymbol{b}) = \exp\left(-\frac{\|\boldsymbol{a} - \boldsymbol{b}\|_2^2}{2\, t_k^2} \right), \qquad t_k = r_k \cdot \text{med}, \tag{2}$$

with multiplicative ratios $\{r_k\}_{k=1}^{K} = \{0.1, 0.5, 1.0, 2.0, 5.0\}$ ($K = 5$). Small ratios produce sharp kernels that resolve fine-grained local neighbourhoods, whereas large ratios produce broad kernels that capture the global manifold geometry. The bank therefore provides K complementary similarity measures that the contrastive stage will reconcile.

3.3. Cross-kernel contrastive representation learning

Per-kernel embeddings.. For each kernel we attach an independent, bias-free linear projection head $g_k(\boldsymbol{x}) = \boldsymbol{W}_k \boldsymbol{x}$ with $\boldsymbol{W}_k \in \mathbb{R}^{d \times D}$, followed by ℓ_2 normalization,

$$\boldsymbol{h}_{k,i} = \frac{\boldsymbol{W}_k \boldsymbol{x}_i}{\|\boldsymbol{W}_k \boldsymbol{x}_i\|_2} \in \mathbb{S}^{d-1}, \qquad k = 1, \ldots, K, \tag{3}$$

so that every embedding lives on the unit hypersphere $\mathbb{S}^{d-1}$. Removing the bias keeps the representation centred at the origin, which is consistent with the origin-separating hyperplane assumption of the downstream OC-SVM, while the normalization equalises the scale across kernels.

Cross-kernel positive pairs.. Whereas conventional contrastive learning forms positive pairs through stochastic data augmentation, we treat the *kernel index as the view*: the two embeddings $\boldsymbol{h}_{k,i}$ and $\boldsymbol{h}_{l,i}$ of the *same* sample under *different* kernels constitute a positive pair, and embeddings of distinct samples are negatives. Consider a mini-batch of B normal samples and a kernel pair (k,l). Stacking the two views yields $\boldsymbol{F}^{(kl)} = [\boldsymbol{h}_{k,1},\ldots,\boldsymbol{h}_{k,B},\boldsymbol{h}_{l,1},\ldots,\boldsymbol{h}_{l,B}]^\top \in \mathbb{R}^{2B\times d}$, whose rows are indexed by $m,n \in \{1,\ldots,2B\}$. The pairwise affinity is the *symmetrised kernel response* evaluated on the embeddings,

$$s_{mn}^{(kl)} = \tfrac{1}{2}\big[\kappa_k(\boldsymbol{f}_m,\boldsymbol{f}_n) + \kappa_l(\boldsymbol{f}_m,\boldsymbol{f}_n)\big]. \tag{4}$$

Because the embeddings are unit-norm, the Gaussian response reduces to a monotone function of cosine similarity, $\kappa_k(\boldsymbol{f}_m,\boldsymbol{f}_n) = \exp(-(1-\langle \boldsymbol{f}_m, \boldsymbol{f}_n\rangle)/t_k^2)$, so t_k^2 plays the role of a contrastive temperature that is automatically adapted to the data scale through Eq. (1).

Cross-kernel InfoNCE.. For anchor n, let $\pi(n)$ index its cross-view positive (i.e. $\pi(n) = n+B$ for $n \le B$ and $\pi(n) = n-B$ otherwise). The InfoNCE loss for the pair (k,l) is

$$\mathcal{L}_{\text{con}}^{(kl)} = -\frac{1}{2B}\sum_{n=1}^{2B}\log\frac{\exp(s_{n,\pi(n)}^{(kl)})}{\sum_{m\neq n}\exp(s_{nm}^{(kl)})}, \tag{5}$$

where the denominator excludes the self-similarity term $m=n$. Averaging over all $\binom{K}{2}$ unordered kernel pairs gives the cross-kernel contrastive objective

$$\mathcal{L}_{\text{con}} = \frac{1}{\binom{K}{2}}\sum_{1\le k<l\le K}\mathcal{L}_{\text{con}}^{(kl)}. \tag{6}$$

Minimising Eq. (6) drives the encoder to a *view-invariant* representation in which all kernels agree on the embedding of each normal sample, i.e. $\boldsymbol{h}_{k,i} \approx \boldsymbol{h}_{l,i}$, while keeping distinct samples mutually discriminable through the negative terms.

3.4. Multi-kernel scatter regularization

Motivation.. The contrastive objective (6) enforces *cross-kernel consistency* but does not constrain the *absolute compactness* of the normal class: it is invariant to a rotation that spreads normal embeddings across the sphere as long as the per-sample views remain aligned. A one-class detector, however, benefits directly from a tightly concentrated normal region. We therefore introduce a regularizer that minimises the dispersion of the normal class within every kernel-induced embedding space, drawing on the one-class specialisation of centroid multiple-kernel k-means (CMKKM).

Definition.. Let $\boldsymbol{\mu}_k = \frac{1}{B}\sum_{i=1}^{B} \boldsymbol{h}_{k,i}$ be the empirical centroid of the normal embeddings under kernel k. The multi-kernel scatter penalty is

$$\mathcal{L}_{\text{scat}} = -\frac{1}{K}\sum_{k=1}^{K} \|\boldsymbol{\mu}_k\|_2^2\,. \tag{7}$$

It is computed in $O(KBd)$ time — a single mean per kernel — and thus adds negligible overhead.

Exact equivalence to within-class scatter.. The next result shows that Eq. (7) is not a heuristic surrogate but is *exactly* (up to an additive constant) the average within-class scatter of the normal embeddings.

Proposition 1 (Scatter equivalence)**.** *For unit-norm embeddings $\boldsymbol{h}_{k,i} \in \mathbb{S}^{d-1}$, define the average normalised within-class scatter $\sigma^2 = \frac{1}{NK}\sum_{k=1}^{K}\sum_{i=1}^{N} \left\|\boldsymbol{h}_{k,i} - \boldsymbol{\mu}_k\right\|_2^2$. Then*

$$\sigma^2 = 1 + \mathcal{L}_{\text{scat}}. \tag{8}$$

Consequently, minimising $\mathcal{L}_{\text{scat}}$ is equivalent to minimising the within-class scatter σ^2, and equivalently to maximising the mean intra-class kernel alignment $\frac{1}{NK}\sum_k \sum_{i,j}\left\langle \boldsymbol{h}_{k,i},\, \boldsymbol{h}_{k,j}\right\rangle = \frac{1}{K}\sum_k \|\boldsymbol{\mu}_k\|_2^2$.

Proof. Fix a kernel k. Expanding the squared norm and using $\left\|\boldsymbol{h}_{k,i}\right\|_2^2 = 1$,

$$\begin{aligned}\sum_{i=1}^{N} \left\|\boldsymbol{h}_{k,i} - \boldsymbol{\mu}_k\right\|_2^2 &= \sum_{i=1}^{N}\Big(\left\|\boldsymbol{h}_{k,i}\right\|_2^2 - 2\,\langle \boldsymbol{h}_{k,i},\, \boldsymbol{\mu}_k\rangle + \|\boldsymbol{\mu}_k\|_2^2 \Big) \\ &= N - 2N\,\|\boldsymbol{\mu}_k\|_2^2 + N\,\|\boldsymbol{\mu}_k\|_2^2 = N(1 - \|\boldsymbol{\mu}_k\|_2^2\,),\end{aligned} \tag{9}$$

where $\sum_i \langle \boldsymbol{h}_{k,i},\, \boldsymbol{\mu}_k\rangle = N\langle \boldsymbol{\mu}_k,\, \boldsymbol{\mu}_k\rangle = N\|\boldsymbol{\mu}_k\|_2^2$ follows from the definition of $\boldsymbol{\mu}_k$. Averaging Eq. (9) over k and dividing by N yields $\sigma^2 = 1 - \frac{1}{K}\sum_k \|\boldsymbol{\mu}_k\|_2^2 = 1 + \mathcal{L}_{\text{scat}}$. The alignment identity is immediate from $\sum_{i,j}\left\langle \boldsymbol{h}_{k,i},\, \boldsymbol{h}_{k,j}\right\rangle = N^2\,\|\boldsymbol{\mu}_k\|_2^2$. □

Remark 1 (Connection to one-class CMKKM)**.** *The one-class instance of centroid multiple-kernel k-means maximises* $\mathrm{Tr}(\boldsymbol{H}^\top \boldsymbol{K}_\eta \boldsymbol{H})$ *with a single soft cluster indicator* $\boldsymbol{H} = N^{-1/2}\mathbf{1}$ *and a convex kernel combination* $\boldsymbol{K}_\eta = \sum_k \eta_k \, \Phi_k \Phi_k^\top$*, where* $\Phi_k = [\boldsymbol{h}_{k,1}, \ldots, \boldsymbol{h}_{k,N}]^\top$ *realises a linear kernel on the embeddings. A short calculation gives* $\mathrm{Tr}(\boldsymbol{H}^\top \boldsymbol{K}_\eta \boldsymbol{H}) = N \sum_k \eta_k \, \|\boldsymbol{\mu}_k\|_2^2$*, which under uniform weights* $\eta_k = 1/K$ *equals* $-N\,\mathcal{L}_{\text{scat}}$*. Hence* $\mathcal{L}_{\text{scat}}$ *is precisely the (negated, scale-normalised) one-class CMKKM compactness objective, providing a multiple-kernel-learning justification for the penalty.*

3.5. Joint objective and collapse avoidance

The encoder $\{\boldsymbol{W}_k\}_{k=1}^K$ is trained by minimising the combined loss

$$\mathcal{L} = \mathcal{L}_{\text{con}} + \lambda\,\mathcal{L}_{\text{scat}}, \tag{10}$$

where $\lambda \geq 0$ trades off cross-kernel consistency against intra-class compactness; $\lambda = 0$ recovers the purely contrastive backbone.

A natural concern is representational *collapse*: the scatter term alone is trivially minimised ($\mathcal{L}_{\text{scat}} = -1$) when all embeddings of a kernel coincide ($\|\boldsymbol{\mu}_k\|_2 = 1$). This degenerate solution is, however, suppressed by the contrastive term. In the alignment–uniformity decomposition of contrastive learning, the numerator of Eq. (5) promotes *alignment* of positives while the log-sum-exp denominator promotes *uniformity* of the embedding distribution on the sphere, which is maximised only by a spread-out configuration and is therefore directly opposed to collapse. The joint objective (10) thus realises a controlled balance: $\mathcal{L}_{\text{scat}}$ pulls each normal class towards its centroid, $\mathcal{L}_{\text{con}}$ prevents the normal embeddings from degenerating to a point and keeps different samples separable, and λ governs the equilibrium. The unit-sphere constraint additionally bounds $\|\boldsymbol{\mu}_k\|_2 \leq 1$, so the penalty cannot diverge.

3.6. Dual-signal anomaly scoring

After training, the projection heads are frozen and each sample $\boldsymbol{x}_i$ is described by *two complementary signals* read off the raw projections $\boldsymbol{W}_k\boldsymbol{x}_i$: the *angular* position of its unit-norm embeddings and the *radial* magnitude of those projections. These signals carry non-redundant discriminative information, and which of them separates the normal and anomalous classes is dataset-dependent. We therefore build a detector for each and fuse their scores.

Directional signal: linear OC-SVM on the angular embedding.. The unit-norm embeddings $\boldsymbol{h}_{k,i}$ retain only the angular position of a sample on each kernel sphere—exactly the geometry that the scatter regularizer (7) compacts. Concatenating the K views,

$$z_i = [\boldsymbol{h}_{1,i}^{\top}, \boldsymbol{h}_{2,i}^{\top}, \ldots, \boldsymbol{h}_{K,i}^{\top}]^{\top} \in \mathbb{R}^{Kd}, \tag{11}$$

fuses them into a single descriptor, and a linear-kernel OC-SVM fitted on the normal embeddings $\{z_i : y_i = 0\}$ by solving

$$\min_{\boldsymbol{w},\rho,\boldsymbol{\xi}} \; \tfrac{1}{2}\,\|\boldsymbol{w}\|_2^2 - \rho + \frac{1}{\nu N_{\mathrm{n}}} \sum_{i:\, y_i=0} \xi_i \quad \text{s.t.} \quad \langle \boldsymbol{w},\, z_i \rangle \geq \rho - \xi_i, \;\; \xi_i \geq 0, \tag{12}$$

yields the *directional score* $s_i^{\mathrm{dir}} = \rho - \langle \boldsymbol{w},\, z_i \rangle$, where $\nu \in (0, 1]$ upper-bounds the fraction of normal training points allowed outside the decision region. A linear separator suffices here because the contrastive and scatter objectives jointly mould the normal class into a compact, nearly linearly separable cone on the product of spheres $(\mathbb{S}^{d-1})^K$; this score is informative whenever the normal/anomaly contrast is *angular*.

Magnitude signal: RBF OC-SVM on the projection norms.. The ℓ_2 normalization in Eq. (3) deliberately discards the projection norms $r_{k,i} = \|\boldsymbol{W}_k \boldsymbol{x}_i\|_2$. Yet these norms are themselves discriminative: trained on normal data alone, each head maps normal samples into a characteristic energy band, whereas anomalies—lying off the learned manifold—are projected to markedly different, typically larger, norms. We collect the per-kernel norms into a magnitude vector

$$\boldsymbol{r}_i = [r_{1,i}, r_{2,i}, \ldots, r_{K,i}]^{\top} \in \mathbb{R}^{K}, \tag{13}$$

and fit an *RBF*-kernel OC-SVM on the normal norms $\{\boldsymbol{r}_i : y_i = 0\}$, producing the *magnitude score* s_i^{mag}. The RBF kernel is essential: a *linear* OC-SVM on norms admits a sign-reversed optimum—when the normal norms are small, the separating direction turns negative and labels large-norm anomalies as inliers—whereas the RBF density model, centred on the compact small-norm normal cluster, correctly assigns high scores to the large-norm outliers lying outside it. This score is informative whenever the contrast is *radial*.

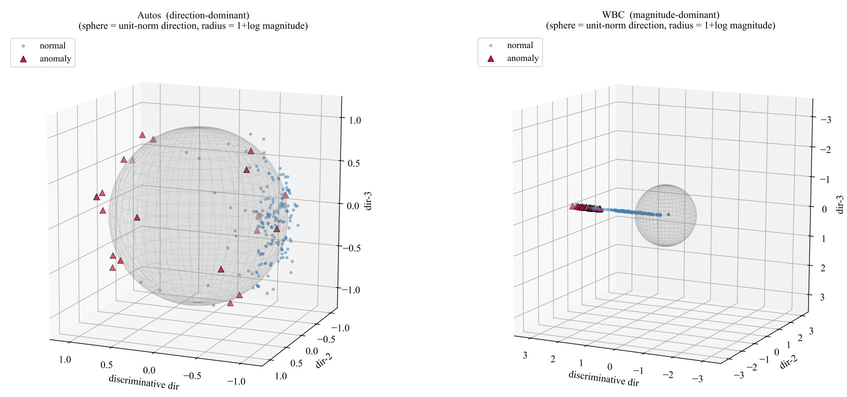
Score fusion.. On a given dataset one signal may be essentially uninformative—most strikingly, the directional score collapses to near-chance when normal and anomalous samples are angularly indistinguishable but differ in projection energy. Without assuming *a priori* which regime holds, we fuse the two detectors by a per-sample maximum of min–max-normalised scores,

$$a(\boldsymbol{x}_i) = \max\Big(\widetilde{s}_i^{\mathrm{dir}},\ \widetilde{s}_i^{\mathrm{mag}}\Big), \qquad \widetilde{s}_i^{\bullet} = \frac{s_i^{\bullet} - \min_j s_j^{\bullet}}{\max_j s_j^{\bullet} - \min_j s_j^{\bullet}}, \tag{14}$$

so that a sample flagged by *either* detector receives a high anomaly score. The fusion needs no prior knowledge of which signal is informative and, as the ablation in Section 4.6 confirms, remains robust on datasets where either single signal collapses while never trailing the stronger signal by a meaningful margin. Fig. 1 makes this complementarity geometric: on a direction-dominant dataset the anomalies stay on the sphere but cluster to one angular region, whereas on a magnitude-dominant dataset they are angularly mixed with the normal class yet bulge far outside the sphere—so a single detector necessarily fails on one regime, while the max-fusion captures both.

3.7. Overall algorithm and complexity

Algorithm 1 summarises the complete training and scoring pipeline. The dominant per-mini-batch cost is the cross-kernel contrastive term, $O(K^2B^2d)$ for the $\binom{K}{2}$ pairwise affinity matrices, while the scatter regularizer contributes only $O(KBd)$. Embedding extraction over the full data set is $O(NKdD)$. Scoring fits two OC-SVMs on the normal set: a linear one on the Kd-dimensional embeddings and an RBF one on the K-dimensional magnitude vectors; both scale with the number of support vectors, and the radial detector is negligible since $K = 5$. With a fixed small kernel bank, the method retains the same asymptotic complexity as the contrastive backbone, so the accuracy gains reported in the experiments come at no additional order-of-growth cost.



(a) Autos — direction-dominant

(b) WBC — magnitude-dominant

Fig. 1. Geometric view of the two complementary signals on the learned representation. The grey sphere is the unit-norm (direction) manifold; a sample's radius is $1 + \log$ of its projection magnitude, so normal points (≈ 1) lie on the shell. **(a)** On Autos the anomalies remain on the shell but concentrate in one angular region—separable by *direction*, while their magnitude is uninformative. **(b)** On WBC the anomalies are angularly intermixed with the normal class yet bulge far outside the shell—separable only by *magnitude*. The angular (linear OC-SVM) detector handles (a), the radial (RBF OC-SVM) detector handles (b), and the max-fusion is robust on both without knowing the regime in advance. For visualization only, the principal in-sphere axis is the normal/anomaly mean-difference direction (labels are used solely for this projection, never for training or scoring).

4. Experiments

This section evaluates whether the theoretical design choices translate into robust detection performance. We first specify the datasets, baselines, metrics and tuning protocol, then compare AUC and G-mean against recent detectors, and finally examine runtime, statistical significance, parameter sensitivity and component ablations.

4.1. Experimental setup

We evaluate on twenty public benchmark datasets[2] drawn from the UCI repository, summarised in Table 1. They span a wide range of sample sizes (101–5171),

[2]https://github.com/BELLoney/Outlier-detection

Algorithm 1 SCMK for one-class anomaly detection

Require: normal set $\mathcal{X}_\mathrm{n}$, full set $\mathcal{X}$, kernel ratios $\{r_k\}_{k=1}^{K}$, embedding dim. d, weight λ, batch size B, epochs T, OC-SVM parameter ν

Ensure: anomaly scores $\{a(\boldsymbol{x}_i)\}_{i=1}^{N}$

med $\leftarrow$ median pairwise distance on $\mathcal{X}_\mathrm{n}$ ▷ Eq. (1)

build kernels κ_k with $t_k = r_k$ med ▷ Eq. (2)

initialise projection heads $\{\boldsymbol{W}_k\}_{k=1}^{K}$

for epoch = 1 **to** T **do**

  for each mini-batch of B samples from $\mathcal{X}_\mathrm{n}$ **do**

    $\boldsymbol{h}_{k,i} \leftarrow \boldsymbol{W}_k \boldsymbol{x}_i / \|\boldsymbol{W}_k \boldsymbol{x}_i\|_2$ for all k, i ▷ Eq. (3)

    $\mathcal{L}_\mathrm{con} \leftarrow$ cross-kernel InfoNCE over kernel pairs ▷ Eqs. (4)–(6)

    $\mathcal{L}_\mathrm{scat} \leftarrow -\frac{1}{K} \sum_k \left\| \frac{1}{B} \sum_i \boldsymbol{h}_{k,i} \right\|_2^2$ ▷ Eq. (7)

    update $\{\boldsymbol{W}_k\}$ by a gradient step on $\mathcal{L}_\mathrm{con} + \lambda \mathcal{L}_\mathrm{scat}$ ▷ Eq. (10)

  end for

end for

$z_i \leftarrow [\boldsymbol{h}_{1,i}; \ldots; \boldsymbol{h}_{K,i}]$, $\boldsymbol{r}_i \leftarrow [\|\boldsymbol{W}_1 \boldsymbol{x}_i\|; \ldots; \|\boldsymbol{W}_K \boldsymbol{x}_i\|]$ for all $\boldsymbol{x}_i \in \mathcal{X}$ ▷ Eqs. (11),(13)

$s^\mathrm{dir} \leftarrow$ linear OC-SVM on $\{z_i : y_i = 0\}$; $s^\mathrm{mag} \leftarrow$ RBF OC-SVM on $\{\boldsymbol{r}_i : y_i = 0\}$ ▷ Eq. (12)

return $a(\boldsymbol{x}_i) = \max\left(\widetilde{s}_i^\mathrm{dir}, \widetilde{s}_i^\mathrm{mag}\right)$ ▷ Eq. (14)

dimensionalities (6–154), anomaly rates (1.9%–25.2%), scenarios, and cover numerical, nominal and mixed-type attributes, so that robustness can be assessed across heterogeneous regimes. Following the protocol adopted in recent anomaly-detection studies, each dataset is split so that 50% of the normal samples form the training set while the remaining 50% of normal samples together with all anomalies form the test set. For the one-class methods—SCMK, Disent-AD, DeepSVDD, LMKAD, ICL and NeuTraLAD—we repeat it for three random seeds $(0, 1, 2)$, training on the training set and scoring on the test set, and report the mean$_{\pm\mathrm{std}}$ AUC and maximum G-mean; KF-GOD and DFNO score samples transductively without a training phase and run on the full set as in their native protocol.

We compare against seven recent outlier detectors: KFGOD [31], Disent-AD [30], DeepSVDD [29], DFNO [28], LMKAD [32], ICL [33] and NeuTraLAD [34]. These include fuzzy-rough/neighbourhood (KFGOD, DFNO), deep one-class representation (Disent-AD, DeepSVDD), multiple-kernel (LMKAD), self-supervised contrastive and transformation (ICL, NeuTraLAD) paradigms, providing a strong and diverse modern reference set.

The encoder of SCMK uses 5 Gaussian kernels with ratios $\{0.1, 0.5, 1, 2, 5\}$ and bias-free linear heads, trained for 100 epochs with Adam (learning rate 0.01, batch size 512) on normal data only; the random seed is fixed to 42. The embedding dimension $d \in \{16, 32, 64, 128, 256\}$, the scatter weight $\lambda \in \{0, 0.1, 1, 10, 100, 1000\}$ and the OC-SVM parameter $\nu \in \{0.01, 0.05, 0.1, 0.2\}$ are selected per dataset. As is customary on these benchmarks, every method—ours and all baselines—is reported at its best hyper-parameter configuration, so the comparison reflects each detector's attainable ceiling under a common protocol.

Detection quality is measured primarily by the area under the ROC curve (AUC), with normal samples labelled 0 and anomalies 1; higher is better. We also report the maximum G-mean, $\max_\tau \sqrt{\mathrm{TPR}(\tau)\mathrm{TNR}(\tau)}$, as a threshold-based complement that balances sensitivity and specificity after a decision threshold is chosen.

4.2. Comparison with recent detectors

Table 2 reports the AUC of SCMK and the seven baselines. SCMK attains the highest mean AUC, $0.914_{\pm 0.022}$, exceeding the strongest single competitor (DFNO, 0.863) by 5.1 points, the next one-class methods LMKAD (0.850) and NeuTraLAD (0.835) by 6.4 and 7.9 points, and the seven-method average (0.824) by 9.0 points; it also has the lowest across-seed standard deviation (0.022) among the one-class methods. SCMK holds the best mean on 9 of the 20 datasets and wins 114 of the 140 pair-wise comparisons. The advantage is clearest where the baselines collectively struggle: on Cardiotocography the best competitor reaches only 0.873 against SCMK's 0.914, on Annealing 0.843 against 0.877, and on Audiology 0.970 against 0.982. On the eleven datasets where SCMK does not hold the best mean it is usually a close second—within 0.01–0.02 of the leader (e.g. Thyroid 0.981 vs. LMKAD/Disent-AD 0.985, Car-

Table 1. The twenty benchmark datasets (N: samples, D: features, ρ: anomaly rate).

Dataset	N	D	ρ	Dataset	N	D	ρ
Vertebral	240	6	12.5%	TicTacToe-69	695	27	9.9%
Thyroid	3772	6	2.5%	WPBC	198	33	23.7%
Ecoli	336	9	2.7%	Ionosphere	249	35	9.6%
Glass	214	9	4.2%	Zoo	101	36	16.8%
WBC	483	9	8.1%	Sick-72	3613	53	2.0%
PageBlocks	5171	10	5.0%	Autos	205	54	12.2%
Wine	129	13	7.8%	Annealing	798	56	5.3%
Cardio	1831	21	9.6%	Lympho.	148	59	4.0%
Cardiotoco.	1688	22	1.9%	Bands-6	318	62	1.9%
TicTacToe-12	638	27	1.9%	Audiology	226	154	25.2%

dio 0.969 vs. LMKAD 0.973, TicTacToe-12 0.990 vs. DFNO 1.000, TicTacToe-69 0.981 vs. DFNO 0.999); the only sizeable shortfalls are Glass and Vertebral, where Disent-AD's deep one-class encoder reaches 0.936 and 0.785. WBC is an angularly indistinguishable case—used to motivate the magnitude branch in Fig. 1—on which SCMK still attains 0.992. That SCMK leads on average across numerical, nominal and mixed-type data, with the lowest across-seed standard deviation (0.022) among the one-class methods, indicates that the kernel-fused, scatter-compacted representation generalises robustly across feature types and random splits rather than exploiting a single data regime. Per-dataset ROC curves for all twenty datasets are given in Fig. 2. The statistical significance of this advantage is established in Section 4.4. As a threshold-based complement to the ranking-based AUC, Table 3 reports the maximum G-mean ($\sqrt{\mathrm{TPR} \cdot \mathrm{TNR}}$) of every method. SCMK again attains the highest average (0.863), ahead of DFNO (0.850), LMKAD (0.834) and NeuTraLAD (0.822), and is best or tied-best on six datasets. The strongest gains occur on difficult low-prevalence and mixed settings such as Ecoli, Cardiotocography, WPBC, Annealing, Lymphography and Audiology, confirming that SCMK's advantage persists once a concrete decision threshold is imposed and is not an artefact of the threshold-free AUC.

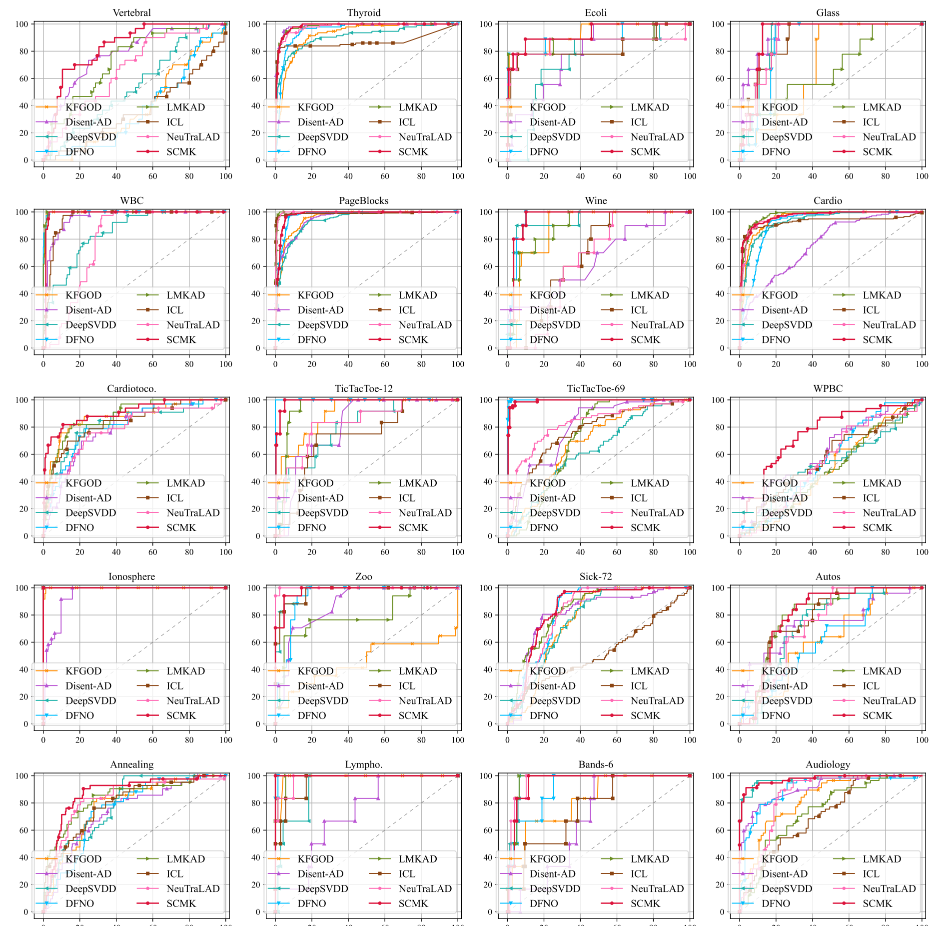


Fig. 2. Per-dataset ROC curves on the twenty datasets.

Table 2. AUC of SCMK against seven recent detectors on twenty datasets. The one-class methods (SCMK, Disent-AD, DeepSVDD, LMKAD, ICL, NeuTraLAD) are reported as $\text{mean}_{\pm\text{std}}$ over three random splits (seeds $0, 1, 2$). Best mean in each row is in **bold**.

Dataset	KFGOD [31]	Disent [30]	DeepSVDD [29]	DFNO [28]	LMKAD [32]	ICL [33]	NeuTraLAD [34]	SCMK (ours)
Vertebral	0.390	**0.785**$_{\pm0.026}$	0.517$_{\pm0.024}$	0.373	0.655$_{\pm0.067}$	0.345$_{\pm0.060}$	0.627$_{\pm0.043}$	0.735$_{\pm0.092}$
Thyroid	0.921	**0.985**$_{\pm0.001}$	0.918$_{\pm0.010}$	0.953	**0.985**$_{\pm0.002}$	0.880$_{\pm0.035}$	0.979$_{\pm0.001}$	0.981$_{\pm0.003}$
WBC	**0.997**	0.964$_{\pm0.004}$	0.864$_{\pm0.024}$	**0.997**	0.992$_{\pm0.003}$	0.967$_{\pm0.008}$	0.808$_{\pm0.034}$	0.992$_{\pm0.007}$
Glass	0.700	**0.936**$_{\pm0.002}$	0.875$_{\pm0.018}$	0.857	0.588$_{\pm0.069}$	0.855$_{\pm0.038}$	0.875$_{\pm0.012}$	0.853$_{\pm0.058}$
Ecoli	0.926	0.834$_{\pm0.041}$	0.761$_{\pm0.042}$	0.905	0.814$_{\pm0.110}$	0.832$_{\pm0.006}$	0.866$_{\pm0.011}$	**0.931**$_{\pm0.010}$
PageBlocks	0.957	0.937$_{\pm0.010}$	0.945$_{\pm0.019}$	0.975	0.995$_{\pm0.001}$	0.989$_{\pm0.004}$	**0.996**$_{\pm0.001}$	0.976$_{\pm0.004}$
Wine	0.887	0.633$_{\pm0.026}$	0.948$_{\pm0.015}$	**0.957**	0.905$_{\pm0.011}$	0.794$_{\pm0.122}$	0.639$_{\pm0.025}$	0.920$_{\pm0.037}$
Cardio	0.950	0.744$_{\pm0.018}$	0.935$_{\pm0.016}$	0.914	**0.973**$_{\pm0.003}$	0.896$_{\pm0.029}$	0.960$_{\pm0.003}$	0.969$_{\pm0.002}$
Cardiotoco.	0.870	0.804$_{\pm0.013}$	0.839$_{\pm0.013}$	0.809	0.873$_{\pm0.014}$	0.751$_{\pm0.102}$	0.754$_{\pm0.024}$	**0.914**$_{\pm0.008}$
TicTacToe-12	0.898	0.811$_{\pm0.111}$	0.750$_{\pm0.043}$	**1.000**	0.938$_{\pm0.004}$	0.694$_{\pm0.067}$	0.823$_{\pm0.009}$	0.990$_{\pm0.003}$
TicTacToe-69	0.690	0.766$_{\pm0.020}$	0.626$_{\pm0.028}$	**0.999**	0.707$_{\pm0.037}$	0.739$_{\pm0.042}$	0.815$_{\pm0.010}$	0.981$_{\pm0.018}$
WPBC	0.494	0.631$_{\pm0.008}$	0.532$_{\pm0.004}$	0.556	0.470$_{\pm0.016}$	0.564$_{\pm0.039}$	0.478$_{\pm0.070}$	**0.653**$_{\pm0.090}$
Ionosphere	0.999	0.954$_{\pm0.017}$	**1.000**$_{\pm0.001}$	**1.000**	**1.000**$_{\pm0.000}$	**1.000**$_{\pm0.000}$	**1.000**$_{\pm0.000}$	**1.000**$_{\pm0.000}$
Zoo	0.457	0.883$_{\pm0.009}$	0.939$_{\pm0.064}$	0.934	0.842$_{\pm0.047}$	0.959$_{\pm0.040}$	0.968$_{\pm0.050}$	**0.969**$_{\pm0.020}$
Sick-72	0.760	0.815$_{\pm0.009}$	0.774$_{\pm0.038}$	0.833	**0.846**$_{\pm0.049}$	0.582$_{\pm0.174}$	0.818$_{\pm0.003}$	0.828$_{\pm0.016}$
Autos	0.625	0.722$_{\pm0.035}$	0.679$_{\pm0.048}$	0.623	0.804$_{\pm0.021}$	0.725$_{\pm0.083}$	0.707$_{\pm0.049}$	**0.822**$_{\pm0.019}$
Annealing	0.783	0.735$_{\pm0.009}$	0.795$_{\pm0.044}$	0.759	0.842$_{\pm0.018}$	0.758$_{\pm0.062}$	0.843$_{\pm0.016}$	**0.877**$_{\pm0.008}$
Lympho.	0.988	0.745$_{\pm0.017}$	0.931$_{\pm0.037}$	0.998	0.995$_{\pm0.006}$	0.947$_{\pm0.021}$	0.976$_{\pm0.007}$	**0.999**$_{\pm0.001}$
Bands-6	0.822	0.677$_{\pm0.012}$	0.936$_{\pm0.047}$	0.909	0.941$_{\pm0.025}$	0.789$_{\pm0.033}$	**0.953**$_{\pm0.021}$	0.914$_{\pm0.044}$
Audiology	0.831	0.887$_{\pm0.020}$	0.970$_{\pm0.009}$	0.903	0.831$_{\pm0.079}$	0.681$_{\pm0.039}$	0.819$_{\pm0.074}$	**0.982**$_{\pm0.008}$
Average	0.797	0.812$_{\pm0.020}$	0.827$_{\pm0.027}$	0.863	0.850$_{\pm0.029}$	0.787$_{\pm0.050}$	0.835$_{\pm0.023}$	**0.914**$_{\pm0.022}$

4.3. Runtime evaluation

Because SCMK forms $\binom{K}{2}$ pairwise cross-kernel affinity matrices per mini-batch and maintains K projection heads, its cost warrants an empirical check rather than an asymptotic argument alone. We therefore measure the actual wall-clock *training* and *inference* time of SCMK against the baselines under a unified protocol: the same GPU, the seed-2 split, a single per-dataset configuration, one CUDA warm-up, and the median of repeated inference passes. Training is the fit on the normal training set; inference is scoring the test set. The transductive detectors (KFGOD, DFNO) have no separate training phase, and to isolate the multiple-kernel overhead we also time the single-kernel variant $\text{SCMK}_{K=1}$. KFGOD and LMKAD are timed through their official

Table 3. Maximum G-mean ($\sqrt{\mathrm{TPR}\cdot\mathrm{TNR}}$) of SCMK against seven recent detectors on twenty datasets, as a threshold-based complement to the AUC in Table 2. The one-class methods (SCMK, Disent-AD, DeepSVDD, LMKAD, ICL, NeuTraLAD) are reported as mean$_{\pm\mathrm{std}}$ over three random splits (seeds $0, 1, 2$); KFGOD and DFNO are transductive on the full set (single-valued). Best mean in each row is in **bold**.

Dataset	KFGOD	Disent	DeepSVDD	DFNO	LMKAD	ICL	NeuTraLAD	SCMK
Vertebral	0.454	**0.741**$_{\pm 0.022}$	0.533$_{\pm 0.005}$	0.436	0.661$_{\pm 0.047}$	0.423$_{\pm 0.055}$	0.624$_{\pm 0.052}$	0.669$_{\pm 0.106}$
Thyroid	0.864	**0.952**$_{\pm 0.003}$	0.855$_{\pm 0.030}$	0.909	0.947$_{\pm 0.005}$	0.882$_{\pm 0.051}$	0.935$_{\pm 0.002}$	0.936$_{\pm 0.007}$
WBC	**0.992**	0.926$_{\pm 0.015}$	0.797$_{\pm 0.025}$	0.985	0.986$_{\pm 0.004}$	0.937$_{\pm 0.004}$	0.816$_{\pm 0.030}$	0.985$_{\pm 0.003}$
Glass	0.752	0.892$_{\pm 0.025}$	0.896$_{\pm 0.011}$	**0.897**	0.613$_{\pm 0.024}$	0.843$_{\pm 0.044}$	0.889$_{\pm 0.003}$	0.861$_{\pm 0.073}$
Ecoli	0.878	0.798$_{\pm 0.054}$	0.781$_{\pm 0.059}$	0.879	0.821$_{\pm 0.101}$	0.866$_{\pm 0.008}$	0.866$_{\pm 0.018}$	**0.890**$_{\pm 0.024}$
PageBlocks	0.895	0.872$_{\pm 0.009}$	0.874$_{\pm 0.031}$	0.948	**0.987**$_{\pm 0.002}$	0.974$_{\pm 0.005}$	0.979$_{\pm 0.001}$	0.953$_{\pm 0.003}$
Wine	0.879	0.641$_{\pm 0.046}$	0.935$_{\pm 0.012}$	**0.948**	0.857$_{\pm 0.035}$	0.797$_{\pm 0.107}$	0.673$_{\pm 0.018}$	0.720$_{\pm 0.227}$
Cardio	0.890	0.681$_{\pm 0.009}$	0.865$_{\pm 0.017}$	0.879	**0.924**$_{\pm 0.004}$	0.852$_{\pm 0.052}$	0.903$_{\pm 0.008}$	0.910$_{\pm 0.010}$
Cardiotoco.	0.835	0.781$_{\pm 0.036}$	0.793$_{\pm 0.006}$	0.778	0.820$_{\pm 0.009}$	0.716$_{\pm 0.082}$	0.719$_{\pm 0.030}$	**0.836**$_{\pm 0.022}$
TicTacToe-12	0.821	0.795$_{\pm 0.093}$	0.714$_{\pm 0.030}$	**1.000**	0.933$_{\pm 0.004}$	0.710$_{\pm 0.072}$	0.811$_{\pm 0.013}$	0.747$_{\pm 0.267}$
TicTacToe-69	0.677	0.751$_{\pm 0.011}$	0.605$_{\pm 0.020}$	**0.990**	0.714$_{\pm 0.027}$	0.689$_{\pm 0.028}$	0.774$_{\pm 0.015}$	0.965$_{\pm 0.034}$
WPBC	0.539	0.629$_{\pm 0.022}$	0.554$_{\pm 0.012}$	0.561	0.505$_{\pm 0.009}$	0.594$_{\pm 0.051}$	0.517$_{\pm 0.067}$	**0.634**$_{\pm 0.076}$
Ionosphere	0.993	0.903$_{\pm 0.031}$	0.997$_{\pm 0.005}$	**1.000**	**1.000**$_{\pm 0.000}$	0.997$_{\pm 0.005}$	**1.000**$_{\pm 0.000}$	0.996$_{\pm 0.008}$
Zoo	0.529	0.843$_{\pm 0.051}$	0.935$_{\pm 0.047}$	0.911	0.802$_{\pm 0.038}$	0.922$_{\pm 0.039}$	**0.963**$_{\pm 0.033}$	0.913$_{\pm 0.048}$
Sick-72	0.744	0.811$_{\pm 0.005}$	0.726$_{\pm 0.043}$	**0.821**	0.782$_{\pm 0.049}$	0.584$_{\pm 0.171}$	0.799$_{\pm 0.015}$	0.816$_{\pm 0.010}$
Autos	0.624	0.716$_{\pm 0.014}$	0.684$_{\pm 0.046}$	0.613	**0.806**$_{\pm 0.021}$	0.720$_{\pm 0.073}$	0.692$_{\pm 0.059}$	0.780$_{\pm 0.021}$
Annealing	0.768	0.697$_{\pm 0.012}$	0.751$_{\pm 0.025}$	0.743	0.800$_{\pm 0.035}$	0.707$_{\pm 0.049}$	0.792$_{\pm 0.013}$	**0.812**$_{\pm 0.025}$
Lympho.	0.979	0.717$_{\pm 0.018}$	0.919$_{\pm 0.033}$	0.993	0.988$_{\pm 0.015}$	0.915$_{\pm 0.011}$	0.944$_{\pm 0.004}$	**0.995**$_{\pm 0.004}$
Bands-6	0.775	0.692$_{\pm 0.030}$	0.927$_{\pm 0.073}$	0.864	**0.940**$_{\pm 0.029}$	0.763$_{\pm 0.043}$	0.932$_{\pm 0.038}$	0.896$_{\pm 0.045}$
Audiology	0.754	0.825$_{\pm 0.017}$	0.933$_{\pm 0.021}$	0.837	0.784$_{\pm 0.080}$	0.653$_{\pm 0.016}$	0.806$_{\pm 0.066}$	**0.948**$_{\pm 0.022}$
Average	0.782	0.783$_{\pm 0.026}$	0.804$_{\pm 0.028}$	0.850	0.834$_{\pm 0.027}$	0.777$_{\pm 0.048}$	0.822$_{\pm 0.024}$	**0.863**$_{\pm 0.052}$

code; the results are summarised in Fig. 3 (means over the twenty datasets).

The multiple-kernel overhead is real but bounded.. SCMK trains in 3.07 s on average versus 0.23 s for SCMK$_{K=1}$, so the $\binom{K}{2}$ affinity computation with $K = 5$ multiplies training time by roughly 13×. In absolute terms, however, SCMK remains firmly mid-pack among the deep detectors: it is faster than NeuTraLAD (4.80 s) and about 5× faster than ICL (15.25 s), and comparable to Disent-AD (2.66 s), while the lighter DeepSVDD (1.04 s) and the kernel one-class LMKAD (0.35 s) are cheaper. SCMK is thus not an outlier in training cost, and the overhead buys the multi-scale representation that drives its accuracy.

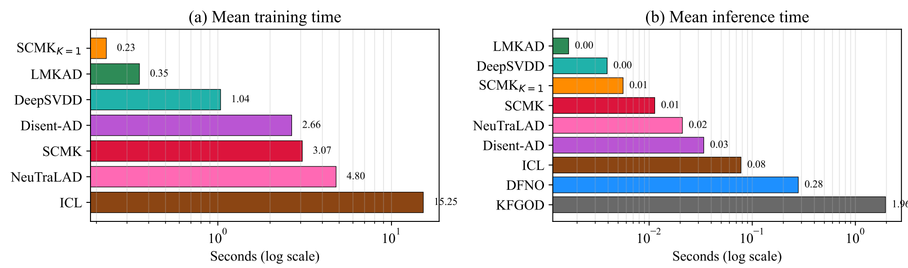


Fig. 3. Mean **(a)** training and **(b)** inference time (seconds, log scale) over the twenty datasets on a single GPU. SCMK$_{K=1}$ is the single-kernel variant; KFGOD and DFNO are transductive and appear only under inference. SCMK's multiple-kernel training cost (3.07 s) stays below NeuTraLAD and ICL, and its inference (11 ms) is among the fastest—far below the transductive KFGOD (1.96 s) and DFNO (0.28 s).

Inference is near-instant and scales gracefully.. At test time SCMK scores a dataset in 11 ms on average—among the fastest methods, since inference is a single forward pass plus two linear/RBF OC-SVM evaluations. The two transductive baselines are orders of magnitude slower and, being $O(n^2)$, degrade sharply with size: on the largest test set (PageBlocks, $n = 2715$) SCMK infers in 0.13 s whereas DFNO takes 2.47 s and KFGOD 27.97 s. SCMK therefore pays a modest one-off training premium for its kernel bank yet is highly efficient to deploy, in contrast to transductive methods that must recompute over the whole data at every query.

4.4. Statistical significance

To confirm that the advantage in Table 2 is statistically significant rather than incidental, we follow the standard protocol of Demšar [27]: a Friedman test across all eight methods (using each method's per-dataset mean AUC), followed by the Nemenyi post-hoc test rendered as a critical-difference (CD) diagram.

The Friedman test strongly rejects the null hypothesis that the eight methods perform equally ($\chi^2_F = 34.98$, $p = 1.1 \times 10^{-5}$; Iman–Davenport $F = 6.33$). Fig. 4 shows the CD diagram at significance level $\alpha = 0.05$ (CD = 2.35). SCMK attains the best mean rank, 2.23, and is separated by more than the critical difference from four of the seven baselines (DeepSVDD 5.25, KFGOD 5.35, DisentAD 5.40, ICL 5.95). The three strongest competitors—LMKAD (3.78), DFNO (3.83) and NeuTraLAD (4.23)—fall

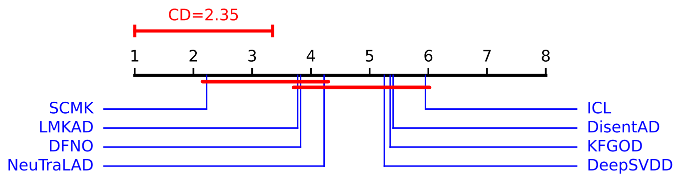


Fig. 4. Critical-difference diagram (Nemenyi post-hoc, $\alpha = 0.05$, CD = 2.35) over the twenty datasets, using each method's per-dataset mean AUC. Lower rank (towards the right) is better; methods not connected by a thick bar are not significantly different. SCMK (mean rank 2.23) has the best rank; DisentAD, DeepSVDD, KFGOD and ICL are significantly worse, while LMKAD, DFNO and NeuTraLAD share its top cluster.

within the critical difference of SCMK, so the conservative Nemenyi test places them in the same top cluster; the pairwise analysis below resolves this finer ordering and still finds SCMK significantly ahead of all three.

We further corroborate this with pairwise one-sided Wilcoxon signed-rank tests of SCMK against each baseline, applying the Holm correction for multiple comparisons (Table 4). SCMK wins against every baseline on a majority of datasets (13–19 of 20), and all Holm-adjusted p-values are below 0.05 (and below 0.01 for six of the seven), so the improvement is significant in all seven comparisons. The margin is largest over the weaker baselines ($p < 10^{-3}$ for KFGOD, DisentAD, DeepSVDD and ICL) and, while smaller, remains significant against the three top-cluster methods that the Nemenyi test could not separate—LMKAD (14/20, Holm $p = 8.9\times10^{-3}$), DFNO (13/20, 1.7×10^{-2}) and NeuTraLAD (16/20, 3.8×10^{-3}).

4.5. Parameter sensitivity

We finally probe how SCMK responds to its three hyper-parameters—the embedding dimension d, the scatter weight λ, and the OC-SVM parameter ν—by varying one at a time on the twenty datasets and averaging the test AUC over the three seeds $\{0, 1, 2\}$. When a parameter is varied, the remaining ones are set to their per-dataset optimum, so each curve reports the AUC attainable at that setting (shaded band: standard deviation across datasets). Fig. 5 summarises the result.

Table 4. Pairwise comparison of SCMK against each baseline (ordered by average rank): mean rank, number of datasets on which SCMK wins, and one-sided Wilcoxon signed-rank p-values before and after Holm correction. ***: $p < 0.001$; **: $p < 0.01$; *: $p < 0.05$.

Method	Avg. rank	SCMK wins	Wilcoxon p	Holm p
LMKAD [32]	3.78	14/20	4.5×10^{-3}	$8.9 \times 10^{-3\,**}$
DFNO [28]	3.83	13/20	1.7×10^{-2}	$1.7 \times 10^{-2\,*}$
NeuTraLAD [34]	4.23	16/20	1.3×10^{-3}	$3.8 \times 10^{-3\,**}$
DeepSVDD [29]	5.25	17/20	6.7×10^{-5}	$3.3 \times 10^{-4\,***}$
KFGOD [31]	5.35	19/20	2.9×10^{-6}	$2.0 \times 10^{-5\,***}$
DisentAD [30]	5.40	17/20	1.6×10^{-4}	$6.4 \times 10^{-4\,***}$
ICL [33]	5.95	18/20	1.3×10^{-5}	$8.0 \times 10^{-5\,***}$
SCMK (ours)	**2.23**	—	—	—

λ is the only sensitive knob.. The scatter weight has by far the widest span (mean AUC ranging over 0.165) and a distinctly non-monotone profile: disabling the penalty ($\lambda = 0$) gives 0.797, a *small* weight is actively harmful (0.724 at $\lambda = 0.1$, 0.742 at $\lambda = 1$), and only once $\lambda \geq 10$ does the term pay off, peaking at 0.890 ($\lambda = 100$) and staying flat to $\lambda = 1000$. A weak scatter penalty perturbs the cross-kernel consistency objective without supplying enough intra-class compactness to compensate; a sufficiently strong one delivers the compactness that one-class detection needs. This corroborates the ablation of Section 4.6 and pins the useful range to $\lambda \in [10, 1000]$.

d and ν are robust.. By contrast, the embedding dimension is nearly flat (span 0.032): any $d \in \{32, \ldots, 256\}$ performs within 0.01 of the $d = 128$ optimum, so SCMK needs no careful width tuning. The OC-SVM ν is likewise stable (span 0.038): AUC is essentially constant for $\nu \leq 0.2$ and declines only gently beyond, favouring a small ν. Overall, SCMK works out-of-the-box for d and ν; the single design choice that matters is giving the scatter term a large enough weight.

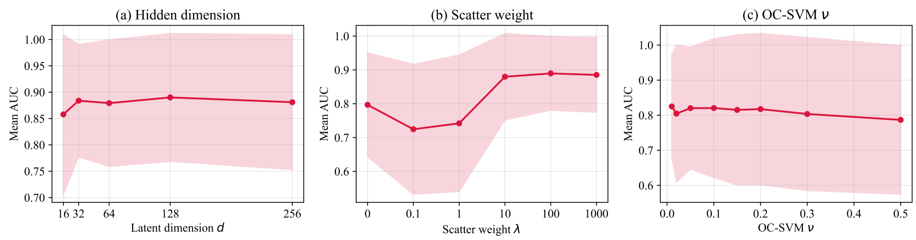


Fig. 5. Parameter sensitivity of SCMK on the twenty datasets (mean test AUC over seeds $\{0, 1, 2\}$; shaded band: standard deviation across datasets). **(a)** Embedding dimension d and **(c)** OC-SVM ν are flat (span 0.03–0.04), so SCMK is insensitive to them. **(b)** The scatter weight λ is the only sensitive parameter: a weak penalty ($\lambda \leq 1$) underperforms even $\lambda = 0$, while $\lambda \geq 10$ is needed for the intra-class compactness that drives detection, plateauing over [10, 1000].

4.6. Ablation study

To isolate the contribution of each component we ablate four design choices on eight representative datasets that deliberately span both detection regimes (angularly separable, e.g. Cardio and Ionosphere; magnitude-separable, e.g. WBC and Autos). The variants are: removing the scatter loss (*w/o* $\mathcal{L}_{\text{scat}}$, i.e. $\lambda = 0$); removing the cross-kernel contrastive loss (*w/o* $\mathcal{L}_{\text{con}}$); replacing the five-kernel bank with a single kernel (*single-kernel*); and restricting the detector to one signal (*dir. only*, the angular score; *mag. only*, the radial score). Table 5 reports the results, and the average row quantifies the mean impact of each removal.

Scatter loss is the dominant component.. Removing the scatter regularizer ($\lambda = 0$) causes by far the largest degradation, −0.206 on average, collapsing Vowels from 0.978 to 0.330 and Bands-6 from 0.957 to 0.517. This confirms the central thesis of the paper: cross-kernel *consistency* alone does not deliver the intra-class *compactness* that one-class detection requires, and the scatter penalty is what supplies it.

Dual-signal fusion is what makes the detector robust.. The two single-signal variants each excel on one regime but *collapse* on the other. The angular detector (*dir. only*) matches the full model on angularly separable data (Cardio 0.967, Ionosphere 1.000, Sonar 0.978) yet falls to 0.586 on WBC, where normal and malignant cells are nearly indistinguishable in angle; conversely the radial detector (*mag. only*) handles WBC

Table 5. Ablation of the four core components (AUC). Best result in each row is in **bold**. The average row shows that the full model is the most robust overall, even though a single-signal variant occasionally peaks on a dataset of its favoured regime.

Dataset	Full	w/o $\mathcal{L}_{\text{scat}}$	w/o $\mathcal{L}_{\text{con}}$	single-kernel	dir. only	mag. only
Ecoli	0.933	0.874	0.931	0.858	0.902	**0.969**
WBC	0.985	0.981	0.984	0.986	0.586	**0.998**
Vowels	**0.978**	0.330	0.977	0.963	0.909	0.906
Cardio	0.967	0.901	0.844	0.703	**0.967**	0.918
Autos	0.805	0.696	0.802	**0.837**	0.830	0.763
Ionosphere	**1.000**	0.866	**1.000**	**1.000**	**1.000**	0.976
Bands-6	0.957	0.517	0.954	0.920	**0.967**	0.917
Sonar	**0.984**	0.794	0.978	**0.984**	0.978	0.951
Average	**0.951**	0.745	0.934	0.906	0.892	0.925

(0.998) but drops to 0.763 on Autos. The fused detector never collapses and attains the highest mean (0.951) *without* any prior knowledge of which signal is informative. The occasional cases where a single signal slightly exceeds the full model on its own favoured dataset (e.g. *dir. only* on Cardio) are precisely offset by its collapse elsewhere — direct evidence that the fusion, not either signal alone, is the robust choice.

Multi-kernel and contrastive terms.. Reducing the bank to a single kernel costs −0.045 on average but is strongly dataset-dependent: it removes 0.26 AUC on Cardio, where multi-scale structure is essential, while leaving datasets with a single dominant scale almost unaffected. Removing the cross-kernel contrastive loss costs the least on average (−0.017); once the scatter penalty is present, the additional consistency constraint contributes only modestly to accuracy, yet it remains important as the uniformity counterweight that prevents representational collapse (Section 3.5).

5. Conclusion and future work

This section closes the argument by summarising how the proposed components address the compactness and multi-scale similarity issues identified earlier. It then outlines limitations of the current protocol and points to natural extensions in adaptive kernels, contaminated training data and streaming or high-dimensional settings.

This paper presented SCMK, a scatter-regularized cross-multi-kernel contrastive framework for one-class anomaly detection. SCMK reconciles a bank of multi-scale Gaussian kernels through a cross-kernel InfoNCE objective that treats the kernel index as the contrastive view, and it compacts the normal class in every kernel-induced space through a multi-kernel scatter regularizer that we proved to be equivalent, up to an additive constant, to the average within-class scatter and to the one-class instance of centroid multiple-kernel k-means. The learned representation is exploited by a lightweight dual-signal detector that fuses an angular (linear OC-SVM) and a radial (RBF OC-SVM) score, recovering the discriminative information that ℓ_2 normalization alone discards. Across twenty benchmark datasets and seven recent detectors, SCMK attained the highest mean AUC (0.914), the highest average maximum G-mean (0.863) and the best mean rank (2.23), with the improvement confirmed significant by Friedman–Nemenyi and Holm-corrected Wilcoxon tests; ablation and sensitivity studies isolated the scatter term as the dominant component and the dual-signal fusion as the source of robustness, while the runtime analysis showed that the multiple-kernel training premium is bounded and inference stays near-instant.

Several directions remain open. The kernel bank currently uses a fixed median-calibrated Gaussian family; learning the bandwidths, or extending to richer kernel families, could adapt the multi-resolution prior to each dataset. The present protocol assumes an uncontaminated normal training set, so relaxing SCMK to tolerate label noise or a small anomaly contamination during training is a natural next step. Finally, because inference reduces to a forward pass and two OC-SVM evaluations rather than a transductive recomputation, extending SCMK to streaming and high-dimensional (e.g. image or graph) settings is a promising avenue for scaling the method beyond tabular anomaly detection.

References

[1] B. Schölkopf, J. C. Platt, J. Shawe-Taylor, A. J. Smola, R. C. Williamson, *Estimating the support of a high-dimensional distribution*, Neural Computation 13 (7) (2001) 1443–1471.

[2] D. M. J. Tax, R. P. W. Duin, *Support vector data description*, Machine Learning 54 (1) (2004) 45–66.

[3] M. M. Breunig, H.-P. Kriegel, R. T. Ng, J. Sander, *LOF: identifying density-based local outliers*, in: ACM SIGMOD International Conference on Management of Data, 2000, pp. 93–104.

[4] F. T. Liu, K. M. Ting, Z.-H. Zhou, *Isolation forest*, in: IEEE International Conference on Data Mining (ICDM), 2008, pp. 413–422.

[5] V. Chandola, A. Banerjee, V. Kumar, *Anomaly detection: A survey*, ACM Computing Surveys 41 (3) (2009) 1–58.

[6] E. M. Knorr, R. T. Ng, V. Tucakov, *Distance-based outliers: algorithms and applications*, The VLDB Journal 8 (3–4) (2000) 237–253.

[7] C. Zhou, R. C. Paffenroth, *Anomaly detection with robust deep autoencoders*, in: ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2017, pp. 665–674.

[8] L. Ruff, R. A. Vandermeulen, N. Görnitz, L. Deecke, S. A. Siddiqui, A. Binder, E. Müller, M. Kloft, *Deep one-class classification*, in: International Conference on Machine Learning (ICML), 2018, pp. 4393–4402.

[9] I. Golan, R. El-Yaniv, *Deep anomaly detection using geometric transformations*, in: Advances in Neural Information Processing Systems (NeurIPS), 2018, pp. 9758–9769.

[10] L. Bergman, Y. Hoshen, *Classification-based anomaly detection for general data*, in: International Conference on Learning Representations (ICLR), 2020.

[11] J. Tack, S. Mo, J. Jeong, J. Shin, *CSI: novelty detection via contrastive learning on distributionally shifted instances*, in: Advances in Neural Information Processing Systems (NeurIPS), 2020, pp. 11839–11852.

[12] C. Qiu, T. Pfrommer, M. Kloft, S. Mandt, M. Rudolph, *Neural transformation learning for deep anomaly detection beyond images*, in: International Conference on Machine Learning (ICML), 2021, pp. 8703–8714.

[13] G. Pang, C. Shen, L. Cao, A. van den Hengel, *Deep learning for anomaly detection: a review*, ACM Computing Surveys 54 (2) (2021) 1–38.

[14] L. Ruff, J. R. Kauffmann, R. A. Vandermeulen, G. Montavon, W. Samek, M. Kloft, T. G. Dietterich, K.-R. Müller, *A unifying review of deep and shallow anomaly detection*, Proceedings of the IEEE 109 (5) (2021) 756–795.

[15] M. Gönen, E. Alpaydın, *Multiple kernel learning algorithms*, Journal of Machine Learning Research 12 (2011) 2211–2268.

[16] G. R. G. Lanckriet, N. Cristianini, P. Bartlett, L. El Ghaoui, M. I. Jordan, *Learning the kernel matrix with semidefinite programming*, Journal of Machine Learning Research 5 (2004) 27–72.

[17] A. Rakotomamonjy, F. R. Bach, S. Canu, Y. Grandvalet, *SimpleMKL*, Journal of Machine Learning Research 9 (2008) 2491–2521.

[18] M. Gönen, E. Alpaydın, *Localized multiple kernel learning*, in: International Conference on Machine Learning (ICML), 2008, pp. 352–359.

[19] X. Liu, Y. Dou, J. Yin, L. Wang, E. Zhu, *Multiple kernel k-means clustering with matrix-induced regularization*, in: AAAI Conference on Artificial Intelligence, 2016, pp. 1888–1894.

[20] H.-C. Huang, Y.-Y. Chuang, C.-S. Chen, *Multiple kernel fuzzy clustering*, IEEE Transactions on Fuzzy Systems 20 (1) (2012) 120–134.

[21] A. van den Oord, Y. Li, O. Vinyals, *Representation learning with contrastive predictive coding*, arXiv:1807.03748, 2018.

[22] T. Chen, S. Kornblith, M. Norouzi, G. Hinton, *A simple framework for contrastive learning of visual representations*, in: International Conference on Machine Learning (ICML), 2020, pp. 1597–1607.

[23] K. He, H. Fan, Y. Wu, S. Xie, R. Girshick, *Momentum contrast for unsupervised visual representation learning*, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020, pp. 9729–9738.

[24] J.-B. Grill, F. Strub, F. Altché, et al., *Bootstrap your own latent: a new approach to self-supervised learning*, in: Advances in Neural Information Processing Systems (NeurIPS), 2020, pp. 21271–21284.

[25] X. Chen, K. He, *Exploring simple Siamese representation learning*, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2021, pp. 15750–15758.

[26] T. Wang, P. Isola, *Understanding contrastive representation learning through alignment and uniformity on the hypersphere*, in: International Conference on Machine Learning (ICML), 2020, pp. 9929–9939.

[27] J. Demšar, *Statistical comparisons of classifiers over multiple data sets*, Journal of Machine Learning Research 7 (2006) 1–30.

[28] Z. Yuan, P. Hu, H. Chen, Y. Chen, Q. Li, *DFNO: detecting fuzzy neighborhood outliers*, IEEE Transactions on Knowledge and Data Engineering 37 (1) (2025) 200–214.

[29] L. Ruff, R. Vandermeulen, N. Görnitz, L. Deecke, S. A. Siddiqui, A. Binder, E. Müller, M. Kloft, *Deep one-class classification*, in: International Conference on Machine Learning (ICML), 2018, pp. 4393–4402.

[30] J. Ye, Z. Tan, Y. Hu, X. Yang, G. Cheng, K. Huang, *Disentangling tabular data towards better one-class anomaly detection*, in: Proceedings of the AAAI Conference on Artificial Intelligence, 2025.

[31] Y. Wu, S. Wang, H. Chen, D. Peng, Z. Yuan, *Kernelized fuzzy-rough anomaly detection*, IEEE Transactions on Fuzzy Systems 32 (8) (2024) 4285–4298.

[32] C. Gautam, R. Balaji, K. Sudharsan, A. Tiwari, K. Ahuja, *Localized multiple kernel learning for anomaly detection: one-class classification*, Knowledge-Based Systems 165 (2019) 241–252.

[33] T. Shenkar, L. Wolf, *Anomaly detection for tabular data with internal contrastive learning*, in: International Conference on Learning Representations (ICLR), 2022.

[34] C. Qiu, T. Pfrommer, M. Kloft, S. Mandt, M. Rudolph, *Neural transformation learning for deep anomaly detection beyond images*, in: International Conference on Machine Learning (ICML), 2021, pp. 8703–8714.